

NOX PROTOCOL

Staking Protocol

TECHNICAL WHITEPAPER

A comprehensive examination of the NOX staking architecture, reward distribution mechanics, and economic framework governing participant incentives within the NOX ecosystem.

VERSION	DATE	NETWORK	STATUS
1.0.0	March 2026	Ethereum	Production

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01 Executive Summary

The NOX Staking Protocol is an incentive mechanism engineered to reward sustained participation and reinforce the economic foundation of the NOX ecosystem. Through time-weighted rewards combined with NFT-based multipliers, the protocol distributes forty million tokens across a twenty-four month emission period.

40M TOTAL REWARDS	2 YEAR PROGRAM	2.5× MAX BOOST	5% SUPPLY ALLOCATED
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Strategic Objectives

The staking protocol addresses the challenge of aligning participant incentives with long-term protocol health. By offering yields to those who commit tokens to the staking contract, the mechanism reduces circulating supply while rewarding community members who demonstrate conviction in the project's trajectory. The proportional distribution model ensures fairness across all participation levels.

The protocol introduces an NFT amplification system providing enhanced rewards to ZeroState Pass holders. This rewards early supporters while creating enduring utility for these digital assets. The tiered multiplier structure—ranging from 1.25× for single NFT holders to 2.5× for those holding five or more—creates meaningful differentiation without excluding participants who lack NFTs.

Emission Structure

PERIOD	EMISSION	DAILY RATE	SHARE
Year One	28,000,000 NOX	76,712 NOX	70%
Year Two	12,000,000 NOX	32,877 NOX	30%
Total	40,000,000 NOX	—	100%

Technical Foundation

The contract leverages OpenZeppelin's audited library with the Universal Upgradeable Proxy Standard (UUPS) enabling future improvements without requiring position migrations. The accumulator pattern ensures gas-efficient reward calculations regardless of participant count, maintaining constant-time operations even at scale.

ARCHITECTURE

- UUPS Upgradeable Proxy
- OpenZeppelin Security Suite
- Role-Based Access Control
- Emergency Pause Capability
- Reentrancy Protection

DESIGN PRINCIPLES

- Gas-Optimized Operations
- Deterministic Calculations
- Minimal Dependencies
- Full On-Chain Transparency
- Comprehensive Event Logging

Deployed Contracts

All contracts are deployed to Ethereum mainnet and verified on Etherscan for complete transparency. The proxy pattern separates storage from logic, with the proxy address serving as the permanent interaction point.

STAKING CONTRACT (PROXY)	0xa94d6009790ba13597a1e1b7cf4e1531ea513613
IMPLEMENTATION	0x2f9d91b41172d8f0fdff30567dc1f8595c2a2d07
NOX TOKEN	0x9a26c80Be4E060e688d7C23aDdB92cBb5D2C9eCA
	NOX Staking Protocol
ZEROSTATE PASS	

KEY INNOVATION

The combination of time-weighted distribution with NFT-based amplification creates a hybrid model rewarding both capital commitment and early ecosystem participation—a meaningful evolution from conventional staking protocols.

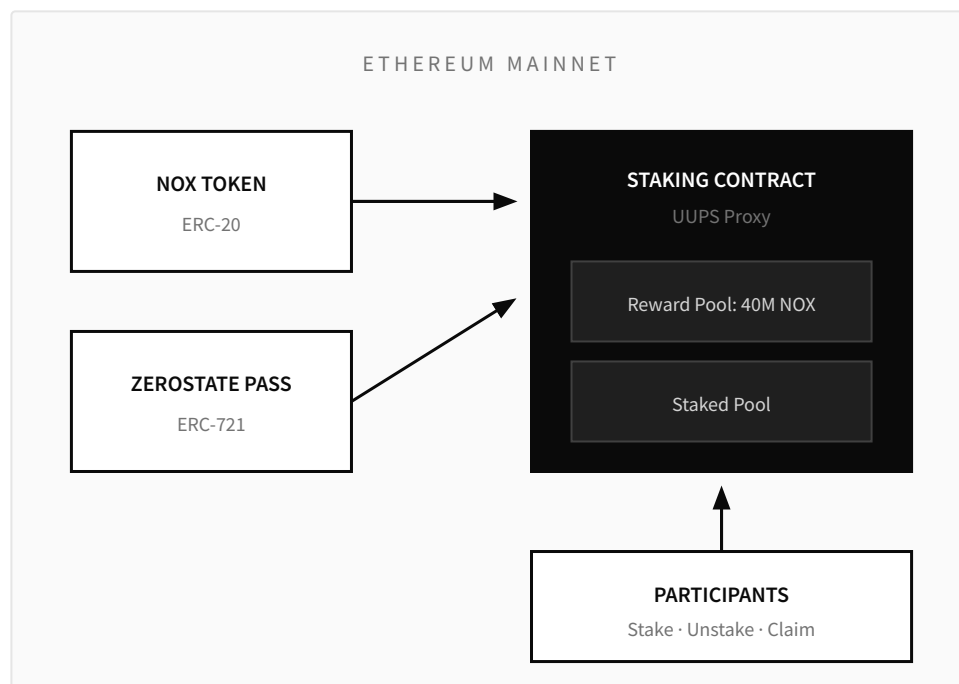
02 Protocol Architecture

The NOX staking protocol operates through a unified smart contract deployed on Ethereum mainnet, interacting with two external token contracts. This section details the structural components and their interactions.

System Overview

The staking contract serves as custodian for two distinct token pools. The staked pool holds user deposits—withdrawable at any time. The reward pool contains forty million NOX allocated for distribution based on proportional stake and duration.

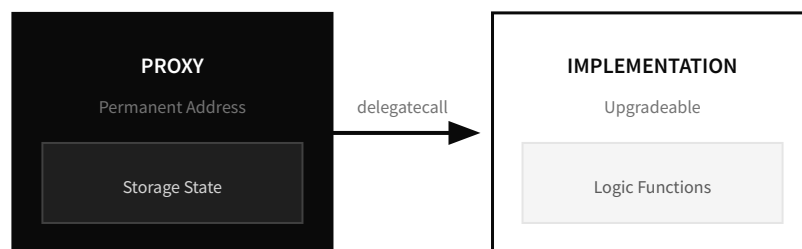
SYSTEM ARCHITECTURE



Proxy Architecture

The Universal Upgradeable Proxy Standard (UUPS) separates storage from logic, enabling protocol updates without position migration. Users interact exclusively with the proxy address, which remains constant. Upgrades modify only the implementation pointer while preserving all storage state.

UUPS PROXY PATTERN



Storage Layout

Contract state is organized into distinct categories: token references, global staking metrics, per-user records, and temporal state. This separation enables efficient queries and maintains upgrade safety.

TOKEN REFERENCES

Immutable pointers to the NOX ERC-20 token and ZeroState Pass ERC-721 collection. These enable deposit operations, withdrawals, reward transfers, and NFT balance queries for boost calculation.

GLOBAL STATE

Aggregate metrics including total staked amount, total weighted stake (with NFT multipliers applied), and the accumulated reward-per-share value. These enable $O(1)$ reward calculations regardless of participant count.

NOX Staking Protocol

USER RECORDS

A mapping from addresses to stake records containing: raw staked amount, weighted amount, reward debt for accumulator math, unclaimed pending rewards, and last interaction timestamp.

TEMPORAL STATE

Genesis timestamp marking emission start, last reward calculation time, and cumulative distributed rewards. These support emission rate determination and protocol analytics.

O4 Staking Operations

The staking mechanism governs how participants deposit tokens, accumulate rewards, and withdraw positions. This section details each operation and the mathematical foundations underlying reward calculation.

Core Principle

Rewards distribute proportionally to weighted stake. A participant controlling ten percent of total weighted stake earns ten percent of emissions during their staking period. No privileged positions or complex tiers exist—larger stakes earn more in absolute terms, but return-per-unit remains equal for all.

$$\text{Reward} = (\text{Weighted Stake} \div \text{Total Weighted}) \times \text{Emission} \times \text{Time}$$

PROPORTIONAL REWARD FORMULA

Stake Operation Sequence

1 Global State Update

Before processing, the contract updates the global accumulator. This ensures existing stakers receive rewards calculated to the current moment, preventing dilution from newly entering stake.

2 Token Transfer

NOX tokens transfer from user wallet to contract via ERC-20 `transferFrom`. Requires prior approval. Reverts on insufficient balance or allowance.

3

NFT Balance Query

The contract queries ZeroState Pass holdings to determine the applicable boost multiplier for this user's stake.

4

Weight Calculation

Weighted stake computed by applying the NFT multiplier. This weighted value determines the user's share of future distributions.

5

State Recording

User record and global totals update. Reward debt set to current accumulator share, establishing the baseline for future reward measurement.

The Accumulator Pattern

Efficient reward calculation at scale requires avoiding per-user iteration. The accumulator pattern achieves constant-time calculation regardless of participant count through mathematical decomposition.

The insight: decompose rewards into a global accumulator tracking total rewards-per-weighted-share, plus per-user debt values tracking each starting point. The difference between current accumulator share and debt yields pending rewards.

```
// Global update (on any state change)
time_elapsed = current_time - last_update_time
new_rewards = time_elapsed * emission_rate
accumulator += (new_rewards * 1e18) ÷ total_weighted_stake
last_update_time = current_time

// User pending calculation
user_share = (user_weighted * accumulator) ÷ 1e18
user_pending = user_share - user_debt + stored_pending
```

The precision constant (10^{18}) maintains numerical accuracy during division. Scaling before division and unscaling after prevents precision loss when dividing small rewards by large stake totals.



Rewards accumulate continuously from stake moment. Unlike epoch-based protocols, NOX calculates per-second, eliminating cliff effects and providing real-time reward visibility.

05 NFT Amplification

The ZeroState Pass collection integration creates synergy between NOX ecosystem asset classes. This section examines NFT-based reward amplification mechanics and strategic implications.

ZeroState Pass

The collection represents founding membership with a maximum supply of 374 pieces. Integration with staking transforms these from pure collectibles into yield-generating instruments with persistent utility.

374 MAX SUPPLY	1.25× SINGLE NFT	2.50× MAXIMUM	5+ FOR MAX BOOST
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Multiplier Structure

The tiered structure rewards accumulation with diminishing marginal returns, encouraging distribution across holders rather than concentration. The jump from four NFTs (2.0×) to five or more (2.5×) creates an exclusive tier for the most committed participants.

NFTS	MULTIPLIER	BASIS POINTS	MARGINAL	100K NOX →	
0	1.00×	10,000	—	100,000	
1	1.25×	12,500	+25%	125,000	16
2	1.50×	15,000	+20%	150,000	

NFTS	MULTIPLIER	BASIS POINTS	MARGINAL	100K NOX →
3	1.75×	17,500	+17%	175,000
4	2.00×	20,000	+14%	200,000
5+	2.50×	25,000	+25%	250,000

Yield Comparison

Practical impact becomes clear through concrete scenarios. Analysis assumes 10M NOX total staked at base 1.0× boost with Year One emissions.

STAKE	NFTS	WEIGHTED	DAILY	ANNUAL
100,000	0	100,000	767	280,000
100,000	1	125,000	959	350,000
100,000	2	150,000	1,151	420,000
100,000	3	175,000	1,342	490,000
100,000	4	200,000	1,534	560,000
100,000	5+	250,000	1,918	700,000

STRATEGIC INSIGHT

A holder with five NFTs and 100,000 staked NOX earns what a non-NFT holder would require 250,000 staked to achieve. This creates significant incentive to acquire ZeroState Pass, supporting secondary market liquidity.

Boost Refresh Mechanics

Multipliers calculate at stake, unstake, or explicit refresh calls. NFT transfers don't auto-update weights—users must trigger refresh to realize changes. The `refreshBoost` function is permissionless, callable by any address for any user.

```
function refreshBoost(address user) external {  
    _updateRewards(user);  
}
```

```
UserStake storage s = stakes[user];
if (s.amount == 0) return;

uint256 newWeighted = calculateWeightedAmount(user, s.amount);

if (newWeighted != s.weightedAmount) {
    totalWeightedStake -= s.weightedAmount;
    totalWeightedStake += newWeighted;
    s.weightedAmount = newWeighted;
    s.rewardDebt = (newWeighted * accRewardPerShare) / 1e18;

    emit BoostUpdated(user, newWeighted);
}
}
```

08 Economic Analysis

Understanding protocol economics enables informed participation decisions. This section examines yield projections, inflation impact, and effects on token dynamics.

APY Projections

Yields vary inversely with participation—more staked tokens means fixed emissions divide among more participants. Tables project APY across total value locked scenarios.

YEAR ONE (28M NOX)

TOTAL STAKED	BASE APY	MAX APY (2.5×)	UTILIZATION
10,000,000	280%	700%	1.25%
25,000,000	112%	280%	3.13%
50,000,000	56%	140%	6.25%
100,000,000	28%	70%	12.50%
200,000,000	14%	35%	25.00%

YEAR TWO (12M NOX)

TOTAL STAKED	BASE APY	MAX APY (2.5×)
10,000,000	120%	300%
50,000,000	24%	60%

TOTAL STAKED	BASE APY	MAX APY (2.5×)
100,000,000	12%	30%
200,000,000	6%	15%

Inflation Analysis

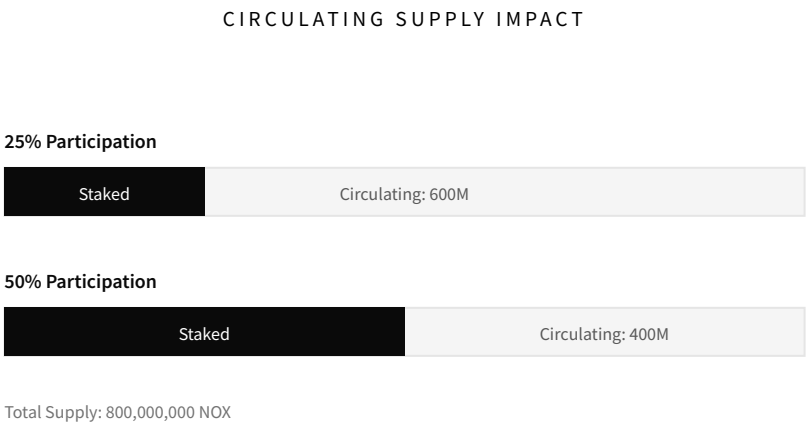
Staking introduces controlled inflation through reward emissions. The total represents five percent of supply over two years, with inflation accruing entirely to active stakers.



Non-stakers experience dilution while active participants may achieve returns significantly exceeding the inflation rate, resulting in net positive outcomes for engaged community members.

Supply Dynamics

Staking removes tokens from active circulation. With 800M total supply, meaningful participation can substantially reduce liquid supply available for trading.



and adoption.

A Risk Considerations

Participation involves risks that prospective stakers should evaluate before committing funds. This section provides an overview of known risk factors.

Smart Contract Risk

Despite rigorous development and audited OpenZeppelin components, contracts may contain undiscovered vulnerabilities potentially resulting in partial or complete fund loss. Upgradeability provides bug-fix capability but introduces its own process risks.

Economic Risk

NOX token value may fluctuate significantly. Staking rewards, while numerically meaningful, may represent reduced purchasing power if prices decline. The fixed emission schedule cannot adapt to market conditions.

Regulatory Risk

Cryptocurrency staking treatment varies across jurisdictions and may change. Participants bear responsibility for compliance with applicable laws. Future regulatory changes could impact protocol operation or reward value.

Technical Risk

The protocol operates on Ethereum and inherits network-level risks including congestion, gas volatility, and potential protocol changes. Operations may become difficult during network stress periods.

This document provides information only and does not constitute financial, legal, or investment advice. Cryptocurrency investments carry substantial loss risk. Conduct independent research and consult qualified professionals before making decisions. Past performance does not guarantee future results.

NOX STAKING PROTOCOL

Technical Documentation v1.0.0

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